

Appendicular Skeleton

BIOL2251
Chapter 8



Axial or
Appendicular



Axial or Appendicular

- **Instructions**

- Work individually
- I will project a list of 12 bones
 - **Sort bones into appendicular (A) or axial (X)**
- Volunteer to sort bones as class

- **Bloom's taxonomy = Remember**

Axial or Appendicular

- Femur
- Sternum
- Humerus
- Lumbar vertebrae
- Clavicle
- Ribs
- Patella
- Temporal bone
- Radius
- Sacrum
- Ilium
- Hyoid

Axial or Appendicular

- Femur = A
- Sternum = X
- Humerus = A
- Lumbar vertebrae = X
- Clavicle = A
- Ribs = X
- Patella = A
- Temporal bone = X
- Radius = A
- Sacrum = X
- Ilium = A
- Hyoid = X



Where Do We Plug In?

Where Do We Plug In?

- **Instructions**

- Work on your own to start
 - **I will project two prompts for you to evaluate**
- Discuss your ideas with a neighbor
- Share as a class

- **Bloom's taxonomy = Remember, Understand**

Where Do We Plug In?

- **Prompt 1:** Name the bones that make up the pectoral girdle. Now describe in your own words where this girdle physically connects to the axial skeleton. Be specific - name the bones involved.
- **Prompt 2:** Name the bones that make up the pelvic girdle. Where does it connect to the axial skeleton? How does this connection differ from the pectoral girdle connection in terms of stability?

Where Do We Plug In?

Pectoral Girdle:

- Bones: Clavicle (collarbone) and scapula (shoulder blade) — one on each side, so 4 bones total
- Axial connection: The clavicle articulates with the manubrium of the sternum at the sternoclavicular (SC) joint — this is the only bony connection between the pectoral girdle and the axial skeleton
- The scapula has no direct bony connection to the axial skeleton; it is held in place entirely by muscles, which allows for exceptional mobility at the expense of stability

Pelvic Girdle:

- Bones: Two hip bones (os coxae), each formed by the fusion of ilium, ischium, and pubis; the two hip bones join anteriorly at the pubic symphysis and posteriorly articulate with the sacrum at the sacroiliac (SI) joints
- Axial connection: Sacroiliac joint — ilium meets the sacrum
- Key contrast: The pelvic girdle connection is far more stable and reinforced compared to the pectoral girdle. The SI joints are tightly bound by some of the strongest ligaments in the body, reflecting the pelvis's role in weight-bearing and locomotion



Name That Bone – Upper Limb Riddles

Name That Bone – Upper Limb Riddles

- **Instructions**

- Work in small groups (12 total)
- I have a list of riddles to read aloud
 - **Each riddle indicates a bone of the upper limbs**
- Write your identifications on the whiteboard
- **Bloom's taxonomy = Remember, Understand**

Name That Bone – Upper Limb Riddles

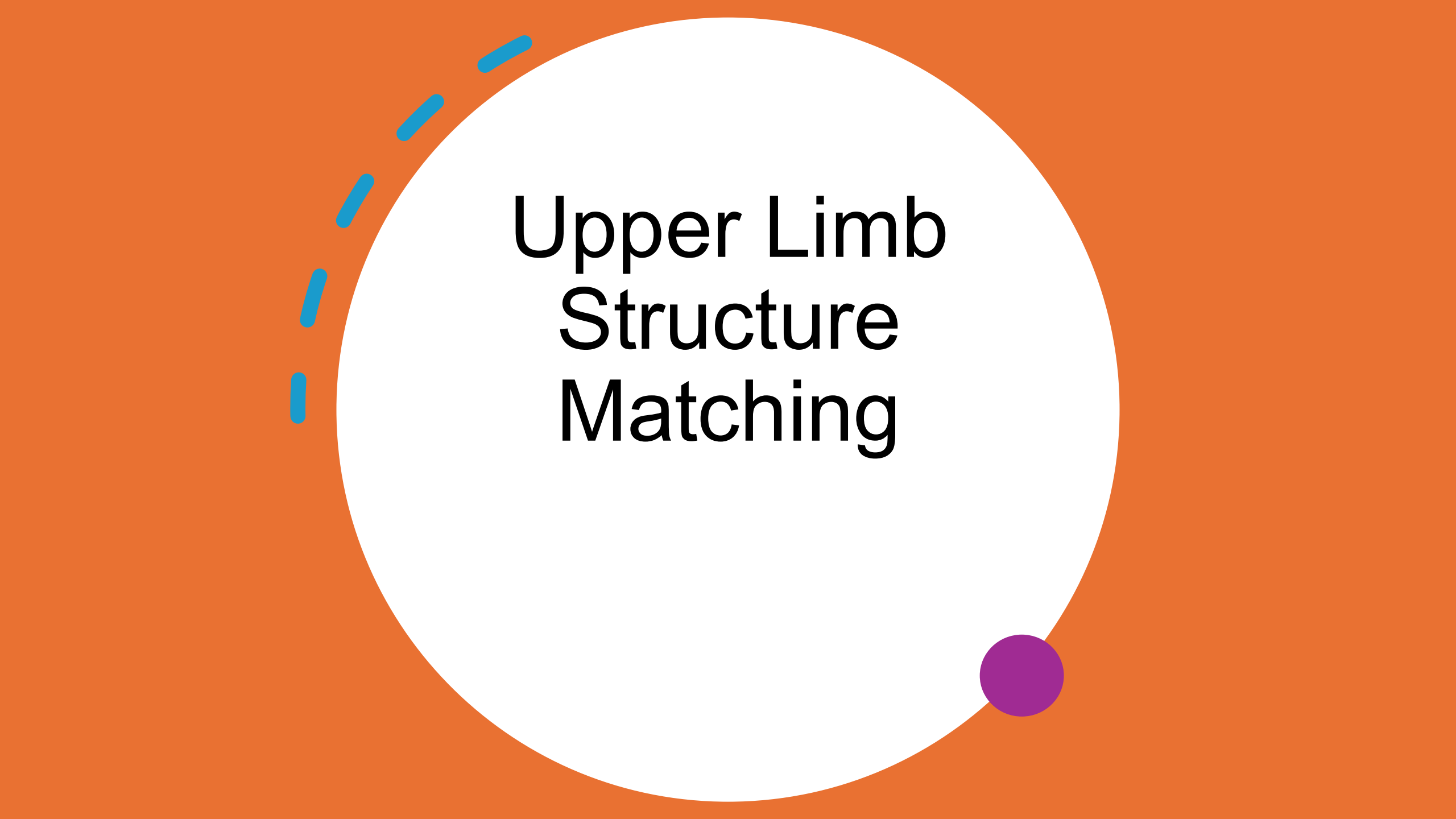
- *Riddle 1:* "I am a flat, triangular bone that sits on your back, not your front. I have no direct bony attachment to the spine or ribs — I float freely, held in place entirely by muscles. I provide a surface for many shoulder muscles to attach. What am I?"
- *Riddle 2:* "I am the single bone of the upper arm. I articulate with a flat triangular bone above me and two forearm bones below me. I have a rounded, ball-like end s that fits into a socket. What am I?"

Name That Bone – Upper Limb Riddles

- *Riddle 3:* "I am on the medial side of the forearm (pinky side). I am the primary bone that articulates with the humerus at the elbow. My proximal end wraps around the humerus like a wrench gripping a bolt. I have a prominent bony bump that you can feel when your elbow is bent — that's my olecranon. What am I?"
- *Riddle 4:* "I am on the lateral side of the forearm (thumb side). My proximal end is small and disc-like, and I don't directly articulate with the humerus in the same way my partner does. I rotate around my partner to pronate and supinate the forearm. My distal end is wider than my proximal end — the opposite of my forearm partner. What am I?"

Name That Bone – Upper Limb Riddles

- Riddle 1 → **Scapula**
- Riddle 2 → **Humerus**
- Riddle 3 → **Ulna**
- Riddle 4 → **Radius**



Upper Limb Structure Matching

Upper Limb Structure Matching

- **Instructions**

- Work in small groups
- I will distribute a table of structure descriptions and name
 - **Match descriptions to the correct structure name**
- I will ask for pairs to volunteer their work
 - Cold-calling if needed
- **Bloom's taxonomy = Remember, Understand**

Upper Limb Structure Matching

- D — Acromion
- E — Glenoid cavity
- F — Coracoid process
- J — Spine (of scapula)
- A — Olecranon fossa
- K — Trochlea
- H — Capitulum
- G — Deltoid tuberosity
- B — Trochlear notch
- L — Olecranon
- M — Radial notch
- C — Radial tuberosity
- I — Styloid process



What's in Your
Hand and Foot?

What's in Your Hand and Foot?

- **Instructions**

- Work on your own before discussing with a partner
 - **Prompt 1 (Hand):** "Using only words — no drawing — describe the skeletal arrangement of the hand, starting at the wrist. How many metacarpals are there? How are they numbered? What are phalanges, how many does each finger have, and how does the thumb differ?"
 - **Prompt 2 (Foot):** "How does the arrangement of bones in the foot parallel the hand? What are the equivalent structures? What is one notable structural difference that reflects the foot's role in weight-bearing versus the hand's role in manipulation?"

- **Bloom's taxonomy = Understand, Apply**

What's in Your Hand and Foot?

Hand:

- The hand contains 5 metacarpals, numbered I–V from the thumb (lateral) to the pinky (medial)
- Each metacarpal has a base (proximal), shaft, and head (distal); the "knuckles" you see are the metacarpal heads
- The phalanges (finger bones) are 14 total: digits II–V each have 3 phalanges (proximal, middle, distal); digit I (thumb) has only 2 (proximal and distal — no middle phalanx)

Foot:

- The foot has 5 metatarsals, numbered I–V from the great toe (medial) to the little toe (lateral) — note that medial/lateral orientation is the reverse of the hand because of limb rotation during development
- Phalanges: 14 total, same formula as the hand — digits II–V have 3 phalanges each; digit I (great toe) has only 2
- Structural difference: Metatarsals and tarsals are arranged to form arches (longitudinal and transverse), which is critical for weight distribution during standing and walking. The hand has no such arch requirement.



Pelvis Face-Off

Pelvis Face-Off

- **Instructions**

- Work in small groups (4-6 students)
- I will provide descriptions of a pelvis
 - **Decide whether pelvis belongs to a biological female or biological male**
 - **Justify your reasoning with at least 3 structural features**
- Groups will share their reasoning
- **Bloom's taxonomy = Analyze, Evaluate**

Pelvis Face-Off

- This pelvis has a broad, fan-shaped ilium
- The angle between the two inferior pubic rami is approximately 100 degrees
- The pelvic inlet (the opening bordered by the pelvic brim) is wide and oval-shaped
- The acetabula are positioned farther apart from each other relative to total pelvis width
- The overall pelvis is shallower and wider

Pelvis Face-Off

Typically female

- Broad, flared ilium → accommodates the wider birth canal
- Subpubic angle $>90^\circ$ (obtuse) → wider outlet for childbirth
- Wide, oval pelvic inlet → again, optimized for fetal passage
- Acetabula spaced wider apart → contributes to the characteristic gait difference between sexes



Name That Bone – Lower Limb Riddles

Name That Bone – Lower Limb Riddles

- **Instructions**

- Work in small groups (12 total)
- I have a list of riddles to read aloud
 - **Each riddle indicates a bone of the lower limbs**
- Write your identifications on the notecard

- **Bloom's taxonomy = Remember, Understand**

Name That Bone – Lower Limb Riddles

- *Riddle 1:* "I am a flat, triangular bone that sits on your back, not your front. I have no direct bony attachment to the spine or ribs — I float freely, held in place entirely by muscles. I provide a surface for many shoulder muscles to attach. What am I?"
- *Riddle 2:* "I am the single bone of the upper arm. I articulate with a flat triangular bone above me and two forearm bones below me. I have a rounded, ball-like end s that fits into a socket. What am I?"

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- *Riddle 4* → **Radius**

Lower Limb Structure Riddles



Lower Limb Structure Riddles

- **Instructions**

- Work on your own
- I have a list of riddles to read aloud
 - **Each riddle indicates a structure on the lower limbs**
- First person with their hand up gets the opportunity to share
- Correct identification (structure + bone) = small prize

- **Bloom's taxonomy = Remember, Understand, Analyze**

Lower Limb Structure Riddles

- "I am a large, rounded projection on the proximal femur, lateral to the neck. I am the attachment point for several powerful hip abductor muscles, including the gluteus medius." → **Greater trochanter (femur)**
- "I am a smaller, knob-like projection located inferior and medial to my larger cousin on the proximal femur. The iliopsoas muscle — the primary hip flexor — attaches to me." → **Lesser trochanter (femur)**
- "I am a deep, central notch on the posterior distal femur, nestled between two large rounded knuckles. I house the cruciate ligaments of the knee." → **Intercondylar fossa (femur)**
- "I am the smooth, central surface on the anterior distal femur. The patella glides over me during flexion and extension of the knee." → **Patellar surface (femur)**

Lower Limb Structure Riddles

- "I am a roughened oval bump on the anterior proximal tibia. The patellar ligament (which connects to the quadriceps) attaches to me, making me the endpoint of the quadriceps mechanism." → **Tibial tuberosity (tibia)**
- "I am the bony projection on the distal medial tibia that forms the inner bump of your ankle. Your deltoid ligament attaches here to stabilize the ankle medially." → **Medial malleolus (tibia)**
- "I am the bony projection at the distal lateral fibula. I form the outer ankle bump. If you sprain your ankle by rolling it outward, the ligaments attaching to me are often damaged." → **Lateral malleolus (fibula)**
- "I am located at the proximal fibula. I am not part of the knee joint itself, but I sit just inferior and lateral to the lateral condyle of the tibia." → **Head (of fibula)**